IPL AUCTION VALUATION AND PLAYER PERFORMANCE INTELLIGENCE

Valuing every IPL auction buy by on-field output per crore, 2013–2026

A Project Report
submitted for the course
BDM (Semester VII)

Submitted by

Abhishek Kokadwar	(2023IMG-002)
Pratik Baviskar	(2023IMG-014)
Rajveer Lomate	(2023IMG-031)
Nabaskar Brahma	(2023IMG-035)



ABV-INDIAN INSTITUTE OF INFORMATION TECHNOLOGY AND MANAGEMENT GWALIOR

GWALIOR, INDIA

OCTOBER, 2026

Abstract

IPL franchises pay at the auction largely for reputation, while matches are won with runs, wickets and catches. This project asks which auction buys earned their price. We combined auction records for 2013–2026 (1,380 buys, ₹3,269 Cr) with ball-by-ball data for 921 matches, scored every player on one Impact Points scale, and built a Value Index: a player's share of league output divided by their share of league spend.

The analysis is delivered as a Power BI dashboard and as a MERN web application backed by MongoDB Atlas. The findings are consistent: prices have doubled since 2013; base-price buys return 4.1× fair value per crore while ₹10 Cr+ signings return 0.49×; and spending is unrelated to winning ($r = -0.03$) while spending efficiently is not ($r = +0.35$).

Keywords: IPL auction, player valuation, Power BI, DAX, MongoDB, data analytics

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1. Introduction

Every year IPL franchises spend hundreds of crores at the player auction. Prices are set in a few minutes of bidding, often driven by reputation, while the return on that money is decided over a season of runs, wickets and catches. Between 2013 and 2026 the franchises spent ₹3,269 Cr on 1,380 auction buys.

Problem statement. Which IPL auction buys earned their price? Taking the view of a franchise analytics team, we measure how much on-field impact each crore bought, so that franchises can be compared and the next auction planned on evidence.

The project answers six business questions:

1. Which franchise gets the most on-field impact per crore spent?
2. Which players were the best bargains and the worst buys?
3. Do franchises that spend efficiently win more matches?
4. Are some roles over- or under-priced, and do overseas players earn their premium?
5. How much have auction prices inflated since 2013?
6. Do base-price buys outperform expensive marquee signings?

The results are delivered in two front ends built on one dataset: a Power BI dashboard (Handover/IPL_Auction_Value.pbix) and a MERN web application with the same pages, which stores the data in MongoDB.

2. Data and Methodology

2.1 Data Sources

Table 2.1: Data sources

Source	What it provides	Size
Kaggle: IPL Auction Dataset 2013–2026	Price, team, role, nationality; base price for 2025–26	1,380 buys
Cricsheet IPL ball-by-ball JSON	Every delivery, fielders, playing XIs, results	921 matches (2013–2026)
Cricsheet Register	Unique player IDs and name variants	All players
Kaggle 2013–22 list; official 2025 list	Missing roles and Indian/Overseas flags	Supplementary
Wisden: IPL 2026 unsold list	Players unsold at the latest auction	79 players

2.2 Data Preparation

All cleaning is done in Power Query inside Power BI, so one Refresh rebuilds the model from the raw files. The main steps were:

- **Standardising auction files:** each year has different columns and price formats; all prices were converted to crores (2013 from US dollars at ₹54 per $).
- **Matching player names:** "Virat Kohli" in the auction is "V Kohli" in Cricsheet. Six matching passes (manual fixes, exact, initials, surname, fuzzy) link 99.0% of auction spend to a player.
- **Team renames:** e.g. Delhi Daredevils → Delhi Capitals, Kings XI Punjab → Punjab Kings; 13 team keys in total.
- **Contracts:** a price covers the auction season and later seasons with the same team, until the next auction or mega auction.

2.3 Data Model

The model is a star schema with three fact tables (player-season, auction buy, team-match) and four dimensions (player, team, season, team-season), all relationships one-to-many. The 93 measures are written in DAX.

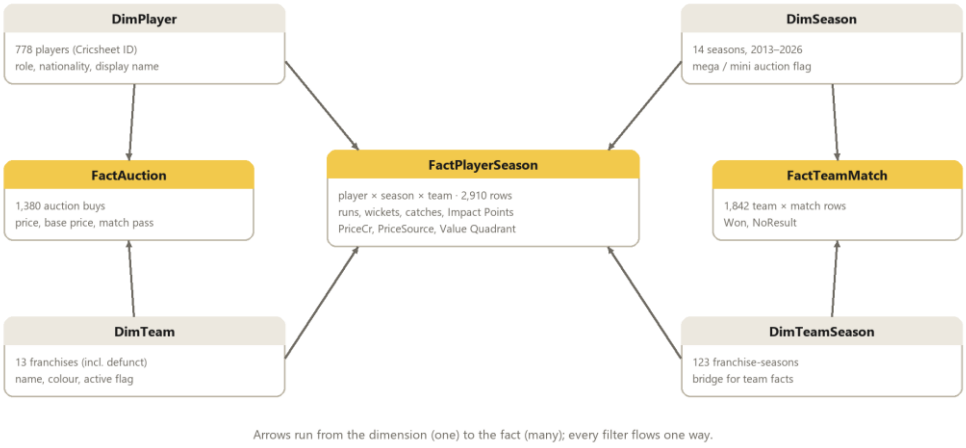


Figure 2.1: Star schema of the Power BI model

2.4 Value Metrics

Impact Points put batting, bowling and fielding on one scale (a run is 1 point, a wicket 25, a catch 8):

$$Impact = Runs + Fours + 2{\cdot}Sixes + 25{\cdot}Wickets + Dot\ balls + 8{\cdot}Catches + 12{\cdot}Stumpings + 6{\cdot}Run\text{-}outs$$

Value Index compares a player's share of the league's impact with their share of the league's auction spend in that season. 1.00 is fair value; above 1 is a bargain, below 1 is overpaid. For

groups across seasons the shares are summed season by season, so each buy is judged against its own year's market.

$$\textit{Value Index} = (\textit{Impact} \div \textit{League Impact}) \div (\textit{Price} \div \textit{League Spend})$$

Fair value and overpay: fair value is what the output would have cost at the league's price per Impact Point; Overpay = Price − Fair value, in crores. **Value quadrants:** each buy is compared with the season's median price and median impact and labelled Star (high price, high impact), Steal (low price, high impact), Overpaid (high price, low impact) or Passenger (low price, low impact).

3. Power BI Dashboard

Each page title states its finding with numbers, a "So what" panel states the decision it supports, and colours keep a fixed meaning (green Steal, yellow Star, red Overpaid, grey Passenger). Table 3.1 lists the pages.

Table 3.1: Pages of the Power BI report

Page	Question answered	Main visuals
Home	Overview	KPI strip and page cards
1 Auction Market	How much have prices inflated?	Average price per season, spend by role
2 Franchise Efficiency	Who gets most impact per crore; does it win?	Value Index by team, spend vs win % scatter
3 Player Value Map	Stars, Steals, Overpaid	Price vs Impact scatter with median lines
4 Steals and Overpaid	Best and worst buys	Top-10 tables, Bargain XI, Overpaid XI
5 Role and Nationality	Are roles / overseas players worth it?	Value Index by role, Indian vs overseas
6 Price Brackets	Do cheap buys beat marquee buys?	Value Index and quadrant mix by bracket
7 Player Profile	One player's history (drill-through)	Price and impact by season
8 Methodology	How is value measured?	Definitions and sources

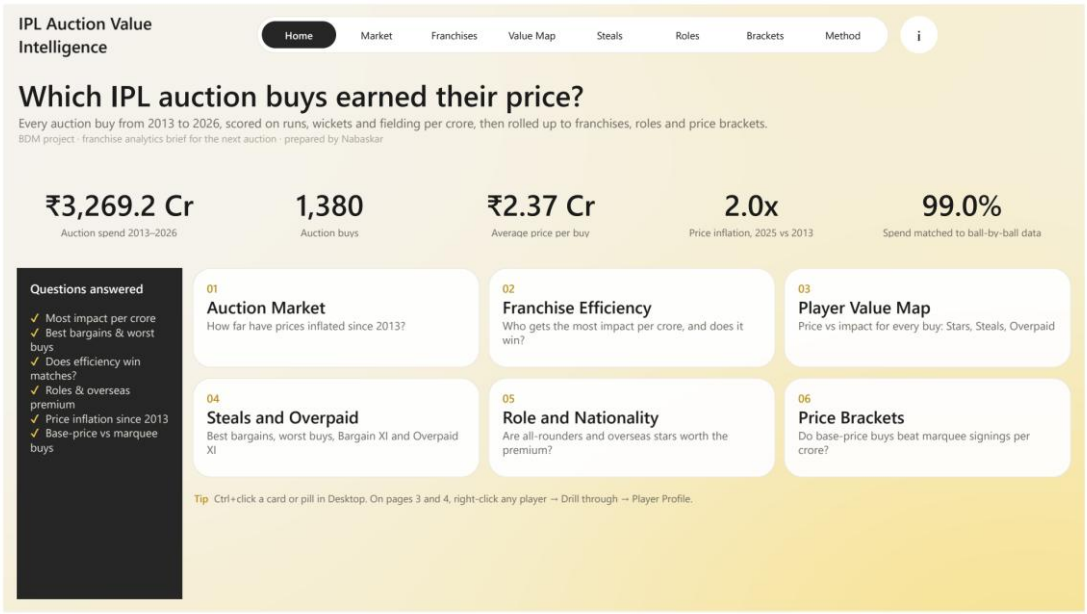


Figure 3.1: Home page

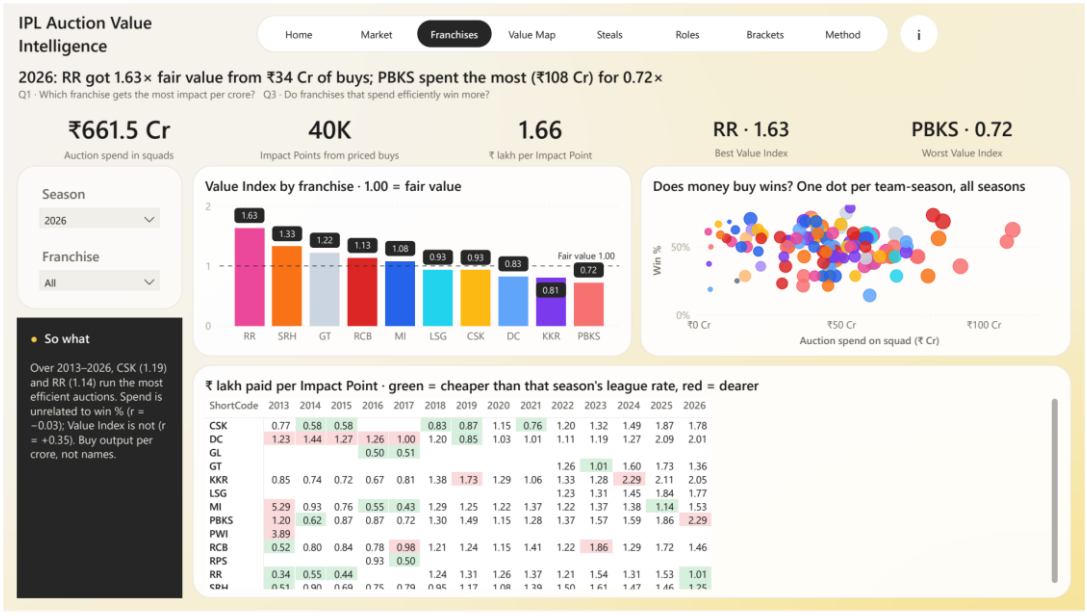


Figure 3.2: Franchise Efficiency

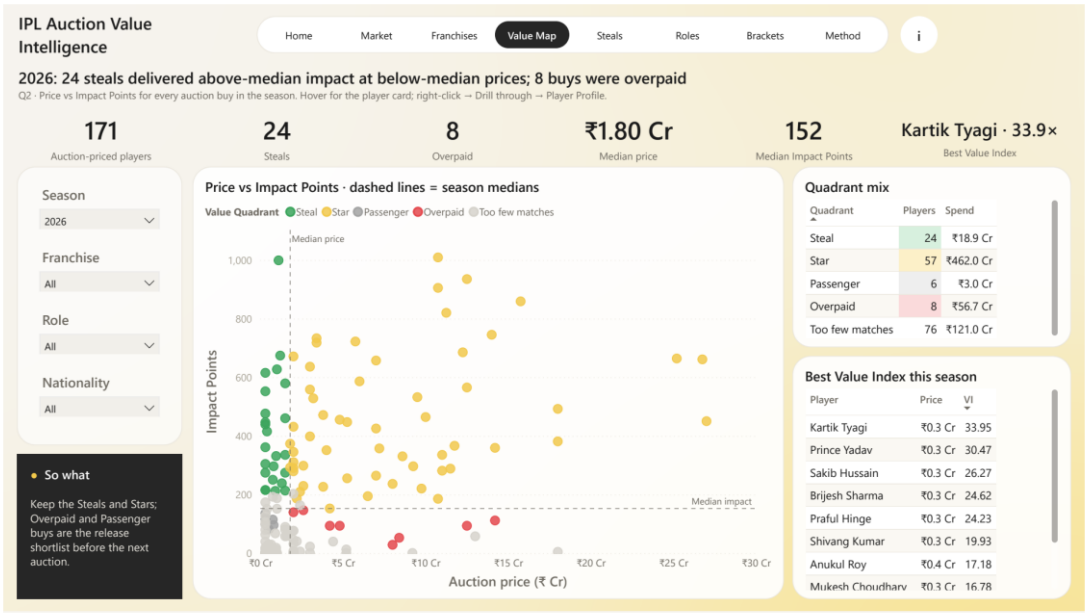


Figure 3.3: Player Value Map

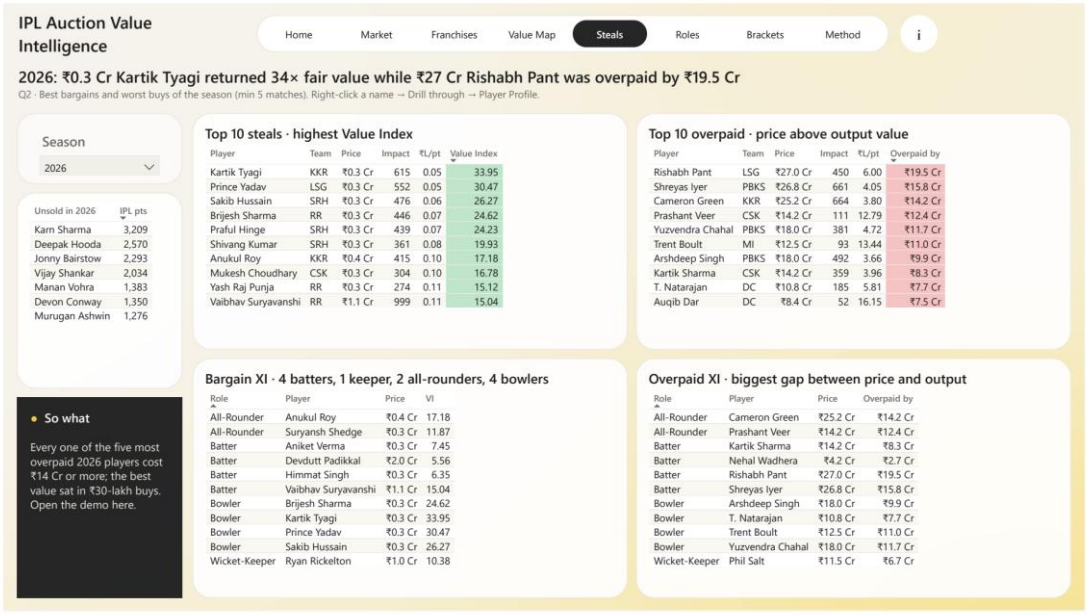


Figure 3.4: Steals and Overpaid

4. Results and Analysis

4.1 Auction Market

Prices have doubled: the average buy cost ₹1.73 Cr in 2013 and ₹3.51 Cr at the 2025 mega auction (2.0×). In 2025–26, bids ended on average 245% above the base price. The highest price is ₹27 Cr for Rishabh Pant (LSG, 2025).

4.2 Franchise Efficiency and Winning

Across 123 franchise-seasons, spend has no link with winning (r = −0.03 between spend and win %), but spending efficiently does (r = +0.35 between Value Index and win %). Over 2013–2026, CSK (1.19) and RR (1.14) ran the most efficient auctions.

Table 4.1: Franchise efficiency in 2026 (top and bottom three)

Franchise	Spend	Win %	Value Index
Rajasthan Royals	₹34.3 Cr	56.3%	1.63
Sunrisers Hyderabad	₹41.8 Cr	60.0%	1.33
Gujarat Titans	₹69.3 Cr	58.8%	1.22
Delhi Capitals	₹73.2 Cr	50.0%	0.83
Kolkata Knight Riders	₹72.8 Cr	46.2%	0.81
Punjab Kings	₹108.1 Cr	53.8%	0.72

4.3 Steals and Overpaid

In 2026, of 171 priced players, 57 were Stars, 24 Steals, 8 Overpaid and 6 Passengers. The best value sat in ₹30-lakh buys, and all five most overpaid players cost ₹14 Cr or more.

Table 4.2: Top steals and most overpaid buys, 2026 (minimum 5 matches)

Top steals	Price	Value Index	Most overpaid	Price	Overpaid by
Kartik Tyagi (KKR)	₹0.3 Cr	33.95	Rishabh Pant (LSG)	₹27.0 Cr	₹19.5 Cr
Prince Yadav (LSG)	₹0.3 Cr	30.47	Shreyas Iyer (PBKS)	₹26.75 Cr	₹15.8 Cr
Sakib Hussain (SRH)	₹0.3 Cr	26.27	Cameron Green (KKR)	₹25.2 Cr	₹14.2 Cr
Brijesh Sharma (RR)	₹0.3 Cr	24.62	Prashant Veer (CSK)	₹14.2 Cr	₹12.4 Cr
Praful Hinge (SRH)	₹0.3 Cr	24.23	Yuzvendra Chahal (PBKS)	₹18.0 Cr	₹11.7 Cr

4.4 Role and Nationality

In 2026 overseas buys cost 2.4× as much as Indian buys but returned less per crore (Value Index 0.88 against 1.11). Over all seasons, Indian bowlers are the best-value role (1.22) and overseas bowlers the worst (0.84).

4.5 Price Brackets

Table 4.3: Value Index by price bracket, 2013–2026

Price bracket	Player-seasons	Spend	Value Index
Base price buy	569	₹189.2 Cr	4.11
Under ₹2 Cr	708	₹592.6 Cr	1.93
₹2 to 5 Cr	494	₹1,531.5 Cr	1.01
₹5 to 10 Cr	271	₹1,868.3 Cr	0.67
₹10 Cr and above	107	₹1,466.6 Cr	0.49

Value per crore falls steadily with price. Marquee players still perform (88% of ₹10 Cr+ player-seasons beat the median impact), but each crore buys far more output in the cheap brackets.

4.6 Validation

Matches per season, Orange and Purple Cap winners and the 2026 auction total (77 players, ₹215.45 Cr) match official records; the league Value Index is exactly 1.00 in every season; and an independent Python re-computation gives identical numbers.

5. Web Application (MERN)

The web app shows the same pages and numbers as the dashboard, so anyone can open it in a browser without Power BI. The only difference is the data layer: the cleaned Power BI tables were exported to JSON and loaded into MongoDB Atlas (8 collections, 7,139 documents). An Express API reads them, React draws the charts, and the site is deployed on Vercel. The web app also adds player search and a Squad Builder that drafts a squad under the ₹120 Cr purse and IPL squad rules.

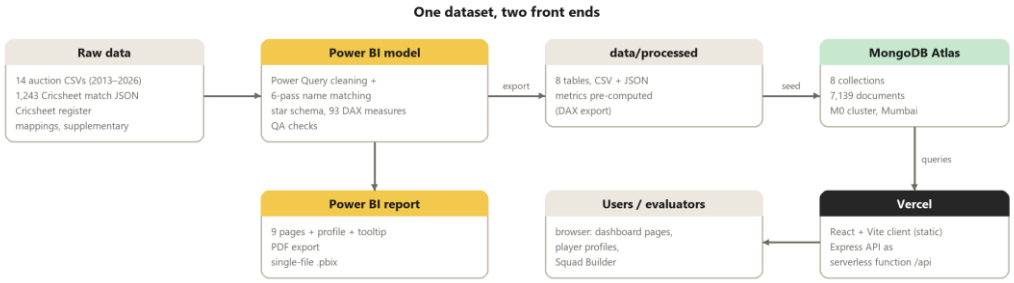


Figure 5.1: Architecture: one dataset, two front ends

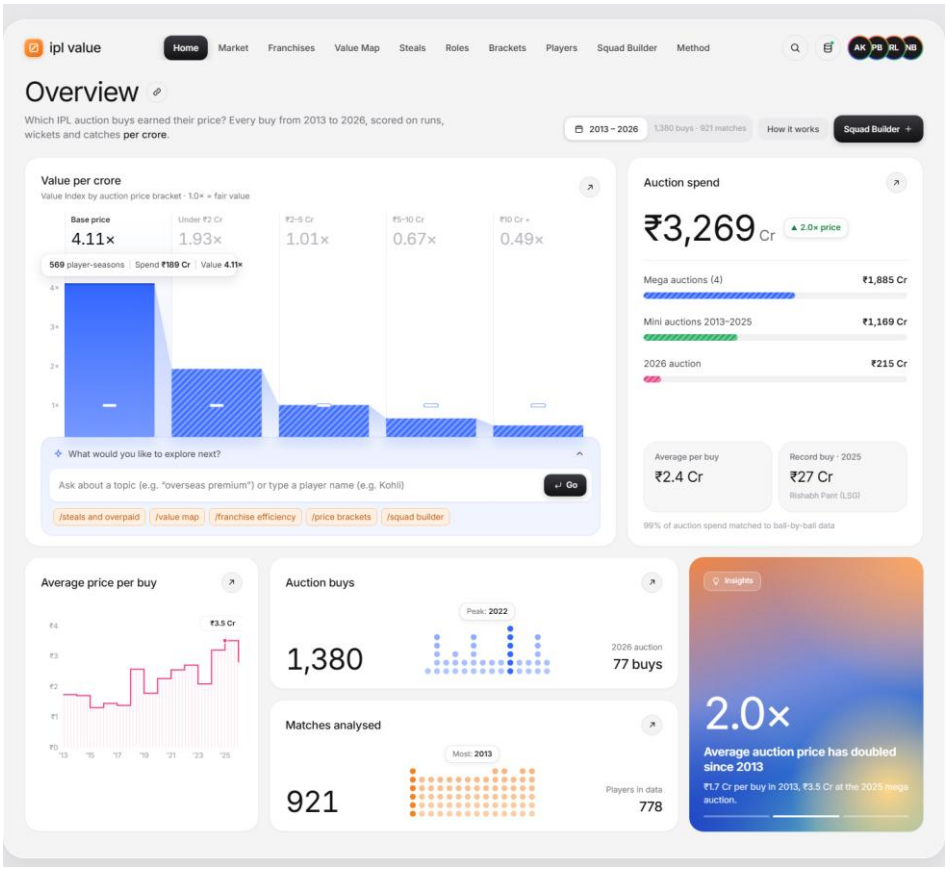


Figure 5.2: Web app Home page

6. Recommendations and Conclusion

6.1 Recommendations

1. **Keep Steals and Stars; release Overpaid and Passengers.** The Value Map gives a ready release shortlist.
2. **Build depth from base-price and under-₹2 Cr buys,** which return 2–4× fair value.
3. **Use overseas slots only on proven match-winners;** the premium is not earned on average.
4. **Cap marquee bids at fair value** and manage to Value Index, not to spend.

6.2 Limitations

- Retained players have no auction price and are excluded from value measures.
- Base prices exist only for 2025–26; Impact Points do not weight match situation.

6.3 Conclusion

By linking 14 years of auction prices to ball-by-ball data and scoring every player on one Impact Points scale, we built a Value Index that compares output with spend fairly across seasons. The answer to our question is consistent: prices have doubled, the most expensive

buys deliver the least per crore, and spending more does not win matches while spending well does. The Power BI dashboard and the MongoDB-backed web app turn this into a practical tool for the next auction.

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Project repository: https://github.com/nabrahma/BDM_Project